

Question No:3

Question description

Alice likes festivals a lot as we all do. He also likes spending money on these festivals. He spends money in buying various things on these festivals. But he has problem of forgetting. He only remembers his top three maximum spendings for any festival.
For eg, on Holi he spends 25 units on colors, 50 units on water sprays, 100 units on gifts, 150 units on sweets but he remebers only his top 3 spendings ,i.e., 50, 100 & 150.
Now as the year ends he wants to know the **festival on which he spent most of his money that he can remember**.

Constraints

- $1 \leq T \leq 100$
- $1 \leq N \leq 10^4$
- $1 \leq X \leq 10^9$
- $1 \leq |S| \leq 10$; $|S|$ denotes string length

S will consist of lowercase and/or uppercase English aplhabets

Input Format

First line contains an integer **T**, denoting number of test cases.
Second line contains an integer **N**, denoting number of spendings in a year.
Each of the next **N** lines contain a string **S** and an integer **X** denoting festival name and spending on one of the events of that festival.

Output Format

For each test case, print a single line containing the festival name and its total spending that he remembers.
If there are more than 1 festival with the maximum spending, print the one which has lexiographically smallest name.

Logical Test Cases

Test Case 1	Test Case 2
INPUT (STDIN) 2 6 B 20 A 2 A 10 A 10 B 30 A 30 3 abc 10 xyz 15 oop 8 EXPECTED OUTPUT A 50 xyz 15	INPUT (STDIN) 2 5 BA 25 AD 24 Ag 10 AE 15 AT 30 3 abc 15 xyz 25 oops 8 EXPECTED OUTPUT AT 30 xyz 25

Code:

```
1  #include <stdio.h>
2  #include <stdlib.h>
3  #include <string.h>
4
5  typedef struct {
6      char name[101];
7      long long total;
8  } Festival;
9
10 int main() {
11     int T;
12     scanf("%d", &T);
13
14     while (T--) {
15         int n;
16         scanf("%d", &n);
17
18         Festival *festivals = malloc(n * sizeof(Festival));
19         int count = 0;
20
21         for (int i = 0; i < n; i++) {
22             char name[101];
23             long long amount;
24
25             scanf("%100s %lld", name, &amount);
26
27             int found = -1;
28
29             for (int j = 0; j < count; j++) {
```

```

30         if (strcmp(festivals[j].name, name) == 0) {
31             found = j;
32             break;
33         }
34     }
35
36     if (found != -1) {
37         festivals[found].total += amount;
38     } else {
39         strcpy(festivals[count].name, name);
40         festivals[count].total = amount;
41         count++;
42     }
43 }
44
45 int best = 0;
46
47 for (int i = 1; i < count; i++) {
48     if (festivals[i].total > festivals[best].total ||
49         (festivals[i].total == festivals[best].total &&
50          strcmp(festivals[i].name, festivals[best].name) < 0)) {
51         best = i;
52     }
53 }
54
55 printf("%s %lld\n",
56        festivals[best].name,
57        festivals[best].total);
58
59 free(festivals);
60 }
61
62 return 0;
63 }

```

Output:

Test Case-1:

```
2
6
B 20
A 2
A 10
A 10
B 30
A 30
A 52
3
abc 10
xyz 15
oop 8
xyz 15
PS C:\Users\kesha\OneDrive\Documents\3s
```

Test Case-2:

```
2
5
BA 25
AD 24
Ag 10
AE 15
AT 30
AT 30
3
abc 15
xyz 25
oop 8
xyz 25
PS C:\Users\kesha\OneDrive\Docu
```

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